

Drawing the Line: How to Detect Gerrymandering through Mathematics

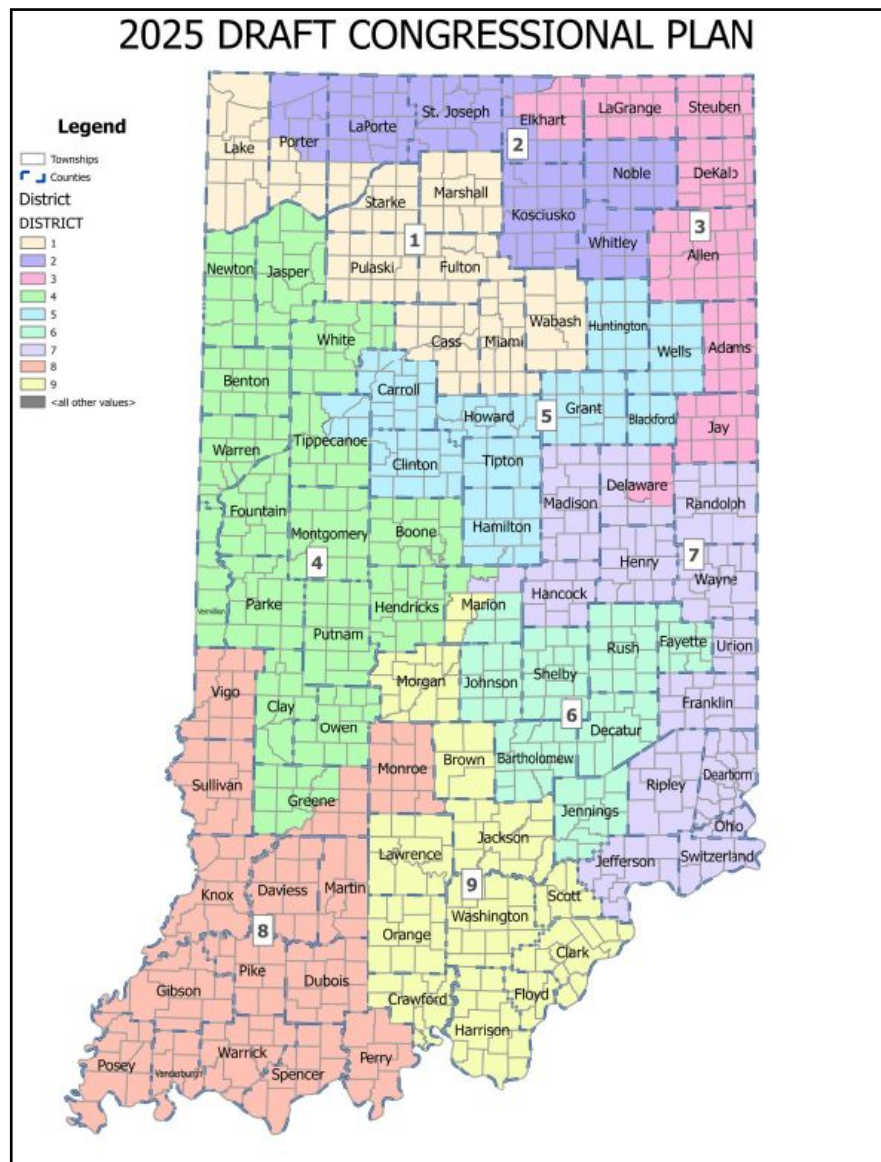
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**“THE MOST EFFECTIVE DEFENSE AGAINST
THE USE OF COMPUTERS TO CREATE
BIASED, PARTISAN MAPS IS THE USE OF
COMPUTERS TO EXPOSE THEM.”**

Abstract

Starting in the summer of 2025, states across the nation took up partisan gerrymandering. A cursory inspection of proposed map in Indiana does not uncover anything amiss. My question became how to detect partisan gerrymandering for a justiciable standard as first introduced by Justice Kennedy in *Vieth v. Jubelirer*. This paper explores many proposed methods, before ending up on recombination as the ideal method.



In November 2025, Indiana state legislators unveiled a redrawn congressional district map that fundamentally altered the state's political landscape. Although Indiana historically maintains a 60-40 Republican voting split, this mid-decade redistricting was strategically modeled to shift the state's 7-2 Republican delegation into a 9-0 congressional sweep. [1] The timing and nature of the announcement—following public pressure from President Donald Trump and lobbying from Vice-President JD Vance, who visited the state multiple times in the lead up to the reveal—bolstered claims by national media that the map was a product of aggressive gerrymandering, the practice of manipulating legislative districts for partisan gain. Indiana Republicans, including Governor Mike Braun and State Representative Andrew Ireland were notably transparent regarding these objectives, with Ireland going so far as to say “Democrats have no business representing us in Washington.” [2] However, a cursory visual inspection of the proposed map does not immediately reveal any manipulation afoot.

Visual hallmarks, such as county splits, compactness, and contiguity have served as the primary metric to detect partisan manipulation since the inception of the practice. In 1812, Massachusetts Governor Elbridge Gerry, a Democratic-Republican, deemed it necessary to massage the electoral lines in order to limit the Federalist party's power in the state senate. The most flagrantly distorted district was located in Essex County; it “meandered around the western and northern county line” to consolidate as many Federalist towns as possible. Capitalizing on this visual absurdity, The Boston Gazette appended “a head, wings, tail, and claws” to this map, christening it a “new species of Monster”: a Gerrymander, a portmanteau of Gerry and salamander, the animal that the district resembled. [3]

In contrast to the 1812 ‘Monster’, the 2025 Indiana map looks remarkably normal. The proposal splits only six of the state's ninety-two counties, and where it does, it largely adheres to township boundaries, the first-level subdivision of counties in Indiana.

Absent are districts that look like an octopus with ‘tentacles’ emanating from a central blob or those districts with ‘holes’ that define typical partisan maps; instead, these districts appear geometrically compact. Moreover, the connectivity of the districts is robust. The intradistrict path between any two points avoids the convoluted routes—such as those traversing bodies of water multiple times or running along the narrow shoulders of a bustling interstate—that have appeared in the past. [4] All things considered, for a map that gives Democrats no business representing Hoosiers in Washington, Indiana’s map looks ordinary.

To understand how a visually compact map can still guarantee a partisan sweep, one must look to the dual mechanics of gerrymandering: packing and cracking. [5] Packing involves concentrating the opposition’s voters into a single district to waste their surplus votes in a lopsided victory; while cracking involves splintering a coherent bloc of voters across multiple districts to ensure they cannot achieve a majority in any of them. The typical strategy is to pack many opposition voters into the same district, and then crack the rest of the voters across the remaining districts. While the original 1812 Gerrymander of Essex county relied on packing the Federalists together, a method easily detected by jagged lines and county breaks, what Indiana has done falls solely under cracking. Some understanding of Indiana political geography would reveal that five of the six county splits target Democratic-leaning urban centers: Porter county around Valparaiso, Elkhart county around Elkhart, Tippecanoe county around West Lafayette, Delaware county around Muncie, and Marion county around Indianapolis. [6] The cracking is most apparent in Marion county, where Indianapolis is partitioned into four separate districts. Consequently, three neighbors in the heart of the state may find themselves represented by three different congressmen, each tied to a district that stretches to the state’s distant western, eastern, or southern borders. By submerging these urban Democratic clusters within sprawling rural Republican constituencies, the map achieved a total partisan sweep without failing the eyeball test.

While the eyeball test provides an intuitive starting point for identifying electoral manipulation, it remains fundamentally subjective. A boundary that appears suspiciously jagged to one observer may seem like a necessary adherence to balance out the population to another. Consequently, the central challenge of determining gerrymandering has historically been establishing an objective, quantifiable standard for partisan gerrymandering that can withstand scrutiny. This search for a mathematical smoking gun was catalyzed by Supreme Court Justice Anthony Kennedy in *Vieth v. Jubelirer* (2004). In his concurring opinion to the majority, Kennedy noted that with “no agreed upon substantive principles of fair districting,” the Court lacked a basis to define “clear, manageable, and politically neutral standards” for measuring how a partisan map burdens representational rights. However, Kennedy left the door ajar, suggesting that the current absence of a workable standard “does not mean that none will emerge in the future.” [7] And since then, scores of mathematicians and political scientists have raced to develop a single definitive metric to enter through that doorway and satisfy the Court’s demand for a neutral arbiter of democratic fairness.

In the race to satisfy Justice Kennedy’s demand for a ‘manageable standard’, early efforts focused on quantifying the visual hallmarks of redistricting: county splits, compactness, and contiguity. The first of these, the frequency of county splits, is a simple counting problem that is easily circumvented by modern mapmakers who are gerrymandering, as evidenced by Indiana’s 2025 map. The last, contiguity, is defined in mathematical terms as path-connected, meaning a traveler can move between any two points within a district without ever leaving it. [8] When geographic anomalies, such as islands are involved, bridges or ferries may serve as this connection. But path-connectedness remains a binary condition rather than a metric, a district is either path-connected or it is not, meaning one can create an extremely complicated district that is only tenuously connected and still pass this test. This leaves compactness as the primary geographic method for detecting gerrymandering. Although thirty-two states have enacted statutes requiring compact districts, their legal definitions range from “doesn’t meander too much” to “those that are square, rectangular, or hexagonal in shape, and not irregularly shaped, to the extent permitted by natural or political boundaries.” [9]

Mathematically, compactness is a functional ratio: a measure of a district's area relative to its perimeter. The underlying theory is that a gerrymandered district will inevitably feature an inflated perimeter that fails to encompass a proportionally significant area.

While several geometric scores exist to quantify compactness, the Polsby-Popper and Reock benchmarks remain the most prominent. The Polsby-Popper score, defined as $PP = 4\pi A/p^2$, calculates the ratio of a district's area (A) to that of a circle with an identical perimeter (P), with a perfect circle representing the ideal of compactness. [10] However, this metric struggles with natural, meandering boundaries; for instance, Indiana's 8th district in its 2021 map follows the Ohio River, resulting in a 'poor' score of .211 despite being graded an 'A' for geographic fairness by the independent Princeton Gerrymandering Project. [11] Conversely, the Reock score measures the ratio between a district's area and the area of the smallest circle that can encompass it. [12] This metric is easily manipulated, as a district can be stripped of specific voting blocs through thin slices and still maintain a high score. Ultimately, whether utilizing Polsby-Popper, Reock, Schwartzberg, Convex Hull, Population Polygon, Length-Width, or any other of the multitude of potential geometric measures, the score will prove insufficient: a map can be fairly drawn yet score poorly, or a high score will hide aggressive gerrymandering. [13] For issues relating to electoral manipulation, maybe an analysis of the votes themselves is needed, instead of an analysis of the map itself.

The leading metric for electoral analysis is the efficiency gap, first proposed in 2015 by Nicholas Stephanopoulos and Eric McGhee. This method quantifies fairness by identifying wasted votes; wasted votes are those cast for a losing candidate or those for a winner that exceed the 50% threshold required for victory. A district where the winning side holds nearly all the wasted votes is considered packed, while one where the losing side holds the wasted votes is competitive (typically seen as a positive). However, when several adjacent districts where most of the wasted votes are on the losing side, it is a sign of a cracked plan. The final efficiency gap is determined by the formula, $EG = (W_A - W_B) / V$, where $W_A - W_B$ is the difference between two parties' wasted votes and V is the total number of votes cast. [14]

Stephanopoulos and McGhee proposed that a value exceeding .08 denotes an unacceptable level of gerrymandering. Their nonprofit organization PlanScore calculated that Indiana's proposed map had a theoretical efficiency gap of .286, [15] highly suggestive of gerrymandering.

Is that it? Is the efficiency gap the smoking gun needed to detect gerrymandering? No. Its mathematical foundation relies on the questionable assumption that a 75-25 vote split is the ideal. As a result, it penalizes any deviation from this ratio. [16] For example, when a party wins 60% of the statewide vote (typical of Indiana), a fair map according to the efficiency gap will give the winning party 80% of the seats, not the 100% that occurs in Indiana's proposed map nor the 60% that proportionality (typically seen as the ideal) dictates. It also tends to struggle with volatility in competitive races. Because a genuinely close election produces lopsided wastage depending on which party gets lucky on election day, a last-minute systematic shift toward one party can make a neutral map look like a deliberate gerrymander. [17] This appears to be the case in Indiana's 2024 data (what PlanScore used for their analysis), where a late surge of undecided voters for Republican candidates skewed the results. [18] Furthermore, while other electoral methods beside the efficiency gap exist, they often rely on the state being competitive (within a 55-45 margin) or both parties winning at least one seat—conditions that Indiana simply does not meet. [19]

The fundamental issue with all of these metrics is their attempt to distill the immense complexity of districting into a singular number. A more effective approach may be seeing what makes a map definitively not gerrymandered. And for that, we return to Massachusetts, a state that has received much criticism from Republicans for its congressional maps. State Representative Andrew Ireland's justification of Indiana's gerrymandering was "[Massachusetts also has nine congressional seats. The political makeup is almost exactly the same [as Indiana], 60/40, but Democrat to Republican. But in Massachusetts, ... it is entirely a Democrat congressional delegation." [20] But the scholarly consensus is that Massachusetts is not partisanly gerrymandered, even though no Republican has represented the state since 1994. [21]

To substantiate this point, a team of researchers at Tufts University generated many quintillion redistricting maps for Massachusetts with total freedom to gerrymander (their only requirement was that the districts were equal in population). Only the 2016 voting data was able to feasibly have proportional representation for Republicans; once the contiguity requirement was restored, no statewide election data since 2000 resulted in a Republican seat. Furthermore, across five separate elections, none of these quintillions of redistricting maps achieved an efficiency gap under 11%, meaning that they were all above the 8% signifying gerrymandering. The data reveals that “[t]hough there are more ways of building a valid districting plan than there are particles in the galaxy, every single one of them would produce a 9–0 Democratic delegation,” meaning that partisan gerrymandering is an impossibility in Massachusetts. [22] This conclusion suggests a new detection strategy for gerrymandering: by comparing a map against a massive ensemble of viable alternatives, one can identify partisan gerrymandering not by a single metric, but whether the map grants a party more seats than the statistical average. The question shifts from ‘Is this map fair?’ to ‘Does this map produce a normal result?’

In order to see if a map is normal, it requires a large collection of viable alternatives, a process that introduces one of the most intriguing challenges in redistricting. The simplest method is the flip-based method, where a computer flips the district of one precinct at a time to create a new temporary map, and after many flips, it creates a new map for the ensemble. It then generates quintillions of these maps. [23] This can be visualized as a blindfolded person standing on a single tile in a large room, taking one random step at a time to explore. After thousands of steps, the person arrives at a new location representing a new map. If repeated enough times, the resulting data points should represent the floor of the entire room. But now imagine if the room was actually two rooms connected by a very long and narrow passageway. The blindfolded traveler may never be able to find the entrance to the corridor, much less navigate to the other side. This same issue arises with the flip-based system, that it may be trapped within a specific neighborhood of maps, failing to represent the entire spectrum of possibilities. The other idea is recombination where two districts are merged and then split along a new boundary.

By ensuring the population remains balanced while allowing the shapes to change drastically, the computer effectively takes much larger leaps. This approach allows the algorithm to traverse the entire space more fully, ensuring the ensemble is an accurate representation of all the potential maps. This approach is also much quicker at generating a large sample, taking only a few hours. [24]

While 21st-century gerrymandering has been revolutionized by initiatives like the Redistricting Majority Project (REDMAP), which leveraged sophisticated software and granular voter data to cement long-lasting partisan advantages, [25] this same computing power now provides a vital neutral framework for evaluating fairness. While these algorithms do not necessarily identify a singular perfect map, they excel at detecting whether a specific proposal offers an extreme, outlier advantage compared to the broader landscape of mathematical possibilities. The most effective defense against the use of computers to create biased, partisan maps is the use of computers to expose them. In the end, the Indiana State Senate voted to reject the proposed map, with every Democrat and a majority of Republicans deeming it gerrymandered and unfit for the state. [26] Instead, the state maintained a map resulting in a 7-2 congressional delegation split; a distribution that aligns with the median outcome seen in the ensemble of Indiana's potential districts. [27]

Notes

- [1] Tom Davies and Leslie Muñiz, “Map Unveiled as Indiana House Gets Down to Redistricting Business,” *Indiana Capital Chronicle*, December 1, 2025, <https://indianacapitalchronicle.com/2025/12/01/house-republicans-start-pushing-indiana-redistricting-map/>
- [2] Steve Inskeep, “Indiana State Rep. Says GOP Must Redistrict to Save Majority,” *NPR*, August 27, 2025, <https://www.npr.org/2025/08/27/nx-s1-5518032/indiana-state-representative-republicansredistricting>.
- [3] Nick Seabrook, *One Person, One Vote: A Surprising History of Gerrymandering in America* (New York: Pantheon Books, 2022), 12–45.
- [4] *Id.*
- [5] Ismar Volić, *Making Democracy Count* (Princeton: Princeton University Press, 2024), 228–272.
- [6] Indiana House Republicans, “2025 Draft Congressional Plan,” November 21, 2025, https://www.indianahousererepublicans.com/clientuploads/2026/Dec._Session/Draft_CD_2025_CountiesTownships.pdf.
- [7] *Vieth v. Jubelirer*, 541 U.S. 267 (2004).
- [8] Volić, *Making Democracy Count*.
- [9] *Id.*
- [10] *Id.*
- [11] Princeton Gerrymandering Project, “Indiana 2021 Congressional - Enacted,” Princeton Gerrymandering Project, October 1, 2023, <https://gerrymander.princeton.edu/redistricting-report-card/?planId=recAzGi2OPAMTkXE8>.
- [12] Volić, *Making Democracy Count*.
- [13] *Id.*

- [14] Mira Bernstein and Moon Duchin, “A Formula Goes to Court: Partisan Gerrymandering and the Efficiency Gap,” *Notices of the American Mathematical Society* 64, no. 9 (2017): 1020–24, <https://doi.org/10.1090/noti1573>.
- [15] PlanScore, “CD_2025 (1).Zip,” PlanScore, December 1, 2025, <https://planscore.org/plan.html?20251201T143828.866082191Z>.
- [16] Bernstein and Duchin, “A Formula Goes to Court.”
- [17] *Id.*
- [18] Marilyn Odendahl, “Republican Wave: Indiana GOP Scores Big Wins in November Election,” *The Indiana Citizen*, November 7, 2024, <https://indianacitizen.org/republican-wave-indiana-gop-scores-big-wins-in-november-election/>.
- [19] PlanScore, “CD_2025 (1).Zip.”
- [20] Inskeep, “Indiana State Rep.”
- [21] Moon Duchin et al., “Locating the Representational Baseline: Republicans in Massachusetts,” *Election Law Journal: Rules, Politics, and Policy* 18, no. 4 (2019): 388–401, <https://doi.org/10.1089/elj.2018.0537>.
- [22] *Id.*
- [23] Daryl DeFord and Moon Duchin, “Random Walks and the Universe of Districting Plans,” in *Political Geometry* (Cham: Springer, 2022), 341–81, https://doi.org/10.1007/978-3-319-69161-9_17.
- [24] *Id.*
- [25] Seabrook, *One Person, One Vote*.
- [26] Brandon Drenon, “Indiana Redistricting: Senate Republicans Side with Democrats to Reject Trump’s Voting Map,” *BBC News*, December 11, 2025, <https://www.bbc.com/news/articles/cgqzzy03p59o>.
- [27] Princeton Gerrymandering Project, “Indiana 2021 Congressional - Enacted.”

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